The Stochastic RALE Model of a Reserve Bank

Mean-Reverting Reserve Dynamics and Balance-Sheet Risk

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Abstract

The original RALE model treats the reserve schedule $R(t)$ as an exogenous, smooth deterministic function. This paper extends the framework by modelling $R(t)$ as an Ornstein–Uhlenbeck (OU) mean-reverting diffusion:

$$\mathrm{d}R = \kappa(\mu_R - R)\,\mathrm{d}t + \sigma\,\mathrm{d}W.$$

At each instant the stochastic reserve level is inserted into the RALE fixed-point equation (equation (4) of the original paper) to recover the full balance-sheet state (A, L, E). We prove that the existence and uniqueness result of Theorem 3.1 carries over *path-by-path* almost surely, derive a delta-method approximation to the stationary distribution of the effective money multiplier, and characterise the probability of insolvency and reserve-floor breach as functions of the OU parameters. Monte Carlo experiments with 500 paths demonstrate solvency-spiral amplification, fat-tailed equity distributions, and the non-trivial interaction between mean-reversion speed κ and balance-sheet volatility.

Keywords: RALE model, Ornstein–Uhlenbeck process, stochastic reserves, money multiplier, insolvency probability, central bank balance sheet.

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1 Introduction

The RALE model introduced a self-referential nonlinear system for the four core balance-sheet quantities of a reserve bank: Reserves $R(t)$, Assets $A(t)$, Liabilities $L(t)$, and Equity $E(t)$. A central assumption of that framework was that $R(t)$ evolves deterministically—either as a smooth exogenous schedule or as the output of a trained machine-learning model. While tractable, this assumption suppresses a fundamental empirical feature: reserve levels at real central banks fluctuate continuously in response to open-market operations, interbank settlement flows, and regulatory buffer adjustments.

This paper replaces the deterministic reserve schedule with a continuous-time stochastic process. We choose the Ornstein–Uhlenbeck (OU) diffusion because it is (i) analytically tractable, (ii) mean-reverting—capturing the tendency of central banks to steer reserves toward a policy target—and (iii) Gaussian, allowing exact simulation without discretisation error.

The paper is organised as follows. Section 2 presents the stochastic extension. Section 3 establishes almost-sure existence and uniqueness. Section 4 derives the stationary multiplier distribution. Section 5 develops risk metrics. Section 6 describes the exact OU discretisation and bisection algorithm. Section 7 reports numerical experiments. Section 8 concludes.

2 The Stochastic RALE Extension

2.1 Reserve Dynamics

We work on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ supporting a standard Brownian motion W. The reserve ratio evolves according to the Itô SDE

$$\mathrm{d}R(t) \;=\; \kappa\big(\mu_R - R(t)\big)\,\mathrm{d}t \;+\; \sigma\,\mathrm{d}W(t), \qquad R(0) = R_0 \in (0,1), \tag{1}$$

where $\kappa > 0$ is the *mean-reversion speed*, $\mu_R \in (0,1)$ is the *long-run policy target*, and $\sigma > 0$ is the *reserve volatility*.

The unique strong solution of (1) is

$$R(t) \;=\; \mu_R + (R_0 - \mu_R)\,e^{-\kappa t} \;+\; \sigma \int_0^t e^{-\kappa(t-s)}\,\mathrm{d}W(s). \tag{2}$$

The conditional distribution of $R(t)$ given $\mathcal{F}_0$ is Gaussian with

$$\mathbb{E}[R(t) \mid \mathcal{F}_0] = \mu_R + (R_0 - \mu_R)e^{-\kappa t}, \qquad \mathrm{Var}[R(t) \mid \mathcal{F}_0] = \frac{\sigma^2}{2\kappa}\big(1 - e^{-2\kappa t}\big). \tag{3}$$

As $t \to \infty$ the marginal distribution converges to the stationary law $\mathcal{N}\big(\mu_R,\, \sigma^2/(2\kappa)\big)$.

2.2 Coupling to the RALE System

At every instant t and on every sample path ω, the reserve level $R(t,\omega)$ is inserted into the RALE fixed-point equation:

$$F(A;\, t,\omega) \;:=\; A \;-\; R(t,\omega)\big[1 + A - E_0(1+A)^t\big] \;=\; 0. \tag{4}$$

Once $A^*(t,\omega)$ is found, the remaining variables follow deterministically:

$$E(t,\omega) = E_0\big(1 + A^*(t,\omega)\big)^t, \quad L(t,\omega) = A^*(t,\omega) - E(t,\omega), \quad m(t,\omega) = \frac{A^*(t,\omega)}{R(t,\omega)}. \tag{5}$$

Because F is strictly monotone in A for any fixed $R \in (0,1)$ (Theorem 3.1 of the original paper), the mapping $R \mapsto A^*(R)$ is well-defined and smooth, so the stochasticity of R propagates into all four state variables.

3 Almost-Sure Existence and Uniqueness

Theorem 3.1 (Stochastic RALE)**.** *Let $R(t)$ be the OU process (1) with $R_0 \in (0,1)$ and let $E_0 \in (0,1)$. Define*

$$\Omega_{\text{good}} \;:=\; \big\{\omega : R(t,\omega) \in (0,1) \text{ for all } t \in [0,T]\big\}.$$

Then $\mathbb{P}(\Omega_{\text{good}}) > 0$ and, on Ω_{good}, there exists a unique $(\mathcal{F}_t)$-adapted process $A^*(t)$ satisfying (4) almost surely.

Proof. On each path $\omega \in \Omega_{\text{good}}$ the trajectory $R(\cdot, \omega)$ is a continuous function valued in $(0, 1)$. Applying Theorem 3.1 of the original paper pointwise yields a unique $A^*(t, \omega)$ for each $t \in [0, T]$.

Measurability. Writing $A^*(t, \omega) = G(R(t, \omega), t)$ where $G : (0, 1) \times [0, T] \to \mathbb{R}$ is defined by the implicit function theorem applied to $F(\,\cdot\,; t, r) = 0$, it suffices to show G is Borel-measurable. Since F is jointly continuous and $\partial F/\partial A > 0$ uniformly, the implicit function theorem guarantees that G is jointly continuous, hence Borel. Composing with the $(\mathcal{F}_t)$-adapted process R yields an $(\mathcal{F}_t)$-adapted process A^*.

Positivity of $\mathbb{P}(\Omega_{\text{good}})$. The OU process exits $(0, 1)$ only when $R(t) \leq 0$ or $R(t) \geq 1$. By (3) and Gaussian tail bounds, the probability that $R(t)$ stays in a compact sub-interval of $(0, 1)$ on $[0, T]$ is strictly positive, so $\mathbb{P}(\Omega_{\text{good}}) > 0$. □

Remark 3.2. For large σ the process $R(t)$ may exit $(0, 1)$ with positive probability. In practice we impose a reflection at $\{10^{-6}, 1-10^{-6}\}$, consistent with the regulatory requirement that reserves remain bounded away from zero and unity.

Corollary 3.3. *Under the hypotheses of Theorem 3.1, the processes $A^*(t)$, $E(t)$, $L(t)$, and $m(t)$ defined by (5) are all $(\mathcal{F}_t)$-adapted and a.s. continuous.*

4 Stationary Distribution of the Money Multiplier

In the long-run stationary regime $R(\infty) \sim \mathcal{N}(\mu_R, \sigma^2/(2\kappa))$ (truncated to $(0, 1)$). The effective money multiplier at a fixed horizon T is $m^*(T) = A^*(T)/R(T)$. Let $g(R) := G(R, T)/R$ denote the multiplier as a function of reserves. Since G is smooth and strictly increasing (Section 3), g is smooth and the change-of-variables formula gives the density of $m^* = g(R)$ as

$$f_{m^*}(x) \;=\; f_R\big(g^{-1}(x)\big) \cdot \left| \frac{\mathrm{d}g^{-1}}{\mathrm{d}x} \right|, \tag{6}$$

where f_R is the (truncated) Gaussian density of the stationary reserve.

Because g is nonlinear, the multiplier distribution is *skewed* even though the reserve distribution is symmetric. A second-order delta-method approximation yields

$$\mathbb{E}[m^*] \;\approx\; g(\mu_R) \;+\; \tfrac{1}{2}\, g''(\mu_R) \cdot \frac{\sigma^2}{2\kappa}, \tag{7}$$

$$\mathrm{Var}[m^*] \;\approx\; \big[g'(\mu_R)\big]^2 \cdot \frac{\sigma^2}{2\kappa}. \tag{8}$$

Equations (7)–(8) show that higher volatility σ or lower mean-reversion speed κ increase both the mean and variance of the effective multiplier, amplifying systemic leverage risk.

5 Risk Metrics

5.1 Insolvency Probability

Define the insolvency event at time t as $\{E(t) \leq \varepsilon_E\}$ for a small threshold $\varepsilon_E > 0$. Since $E(t) = E_0(1 + A^*(t))^t$ is monotone increasing in $A^*(t)$, and $A^*(t)$ is increasing in $R(t)$, insolvency corresponds to sufficiently low reserves:

$$\mathbb{P}\big(E(t) \leq \varepsilon_E\big) \;=\; \mathbb{P}\big(R(t) \leq R^*(t)\big), \tag{9}$$

where $R^*(t)$ is the unique reserve level solving $E_0\big(1 + G(R^*(t), t)\big)^t = \varepsilon_E$. Since $R(t)$ is Gaussian with moments given by (3), we obtain the closed-form expression

$$\mathbb{P}\big(E(t) \leq \varepsilon_E\big) \;=\; \Phi\left(\frac{R^*(t) - \mu_{\text{eff}}(t)}{\sigma_{\text{eff}}(t)}\right), \tag{10}$$

where $\mu_{\text{eff}}(t) = \mu_R + (R_0 - \mu_R)e^{-\kappa t}$, $\sigma_{\text{eff}}(t) = \sigma\sqrt{(1 - e^{-2\kappa t})/(2\kappa)}$, and Φ is the standard normal CDF.

5.2 Reserve-Floor Breach

A reserve-floor breach occurs when $R(t)$ falls below the regulatory minimum R_{floor}. By the same Gaussian structure:

$$\mathbb{P}\big(R(t) \le R_{\text{floor}}\big) \;=\; \Phi\left(\frac{R_{\text{floor}} - \mu_{\text{eff}}(t)}{\sigma_{\text{eff}}(t)}\right). \tag{11}$$

This probability increases with σ and decreases with κ, confirming the intuition that faster mean reversion insulates the balance sheet against liquidity shocks.

6 Exact Discretisation and Algorithm

The OU process admits an *exact* discretisation (no Euler–Maruyama error). For a uniform grid $t_k = k\Delta t$, $k = 0, \ldots, N$, with $\Delta t = T/N$:

$$R(t_{k+1}) \;=\; \mu_R + \big(R(t_k) - \mu_R\big)e^{-\kappa\Delta t} \;+\; \sigma\sqrt{\tfrac{1-e^{-2\kappa\Delta t}}{2\kappa}}\, Z_k, \qquad Z_k \overset{\text{iid}}{\sim} \mathcal{N}(0,1). \tag{12}$$

The full stochastic RALE simulation algorithm is given in Algorithm 1.

Algorithm 1 Stochastic RALE Simulation

Require: OU params $(\kappa, \mu_R, \sigma, R_0)$; initial equity E_0; horizon T; steps N; tolerance ε
Ensure: Paths $\{R(t_k), A^*(t_k), L(t_k), E(t_k)\}_{k=0}^{N}$

```
Pre-compute e^{-κΔt} and σ_disc = σ√((1 − e^{-2κΔt})/(2κ))
Draw Z_0, ..., Z_{N-1} ~iid N(0,1)
Set R(t_0) ← R_0
for k = 0, ..., N − 1 do
    R(t_{k+1}) ← μ_R + (R(t_k) − μ_R)e^{-κΔt} + σ_disc Z_k
    Clip R(t_{k+1}) to (10^{-6}, 1 − 10^{-6})
end for
for k = 0, ..., N do
    A*(t_k) ← Bisection(F(·; t_k, R(t_k)), [−0.999, 50], ε)
    E(t_k) ← E_0 (1 + A*(t_k))^{t_k}
    L(t_k) ← A*(t_k) − E(t_k)
end for
```

Proposition 6.1 (Convergence)**.** *The inner bisection loop in Algorithm 1 requires at most* $\lceil \log_2(50.999/\varepsilon) \rceil$ *iterations. For* $\varepsilon = 10^{-9}$ *this is fewer than* 34 *iterations, independent of the OU path.*

Proof. Follows directly from Proposition 4.1 of the original paper, applied with bracket $[a_0, b_0] = [-0.999, 50]$. □

7 Numerical Experiments

7.1 Baseline Parameterisation

We use $\kappa = 2.0$, $\mu_R = 0.5$, $\sigma = 0.08$, $R_0 = 0.5$, $E_0 = 0.05$, $T = 5$, $N = 200$. Figures 1 and 2 show a representative single path and the Monte Carlo ensemble (500 independent paths), respectively.

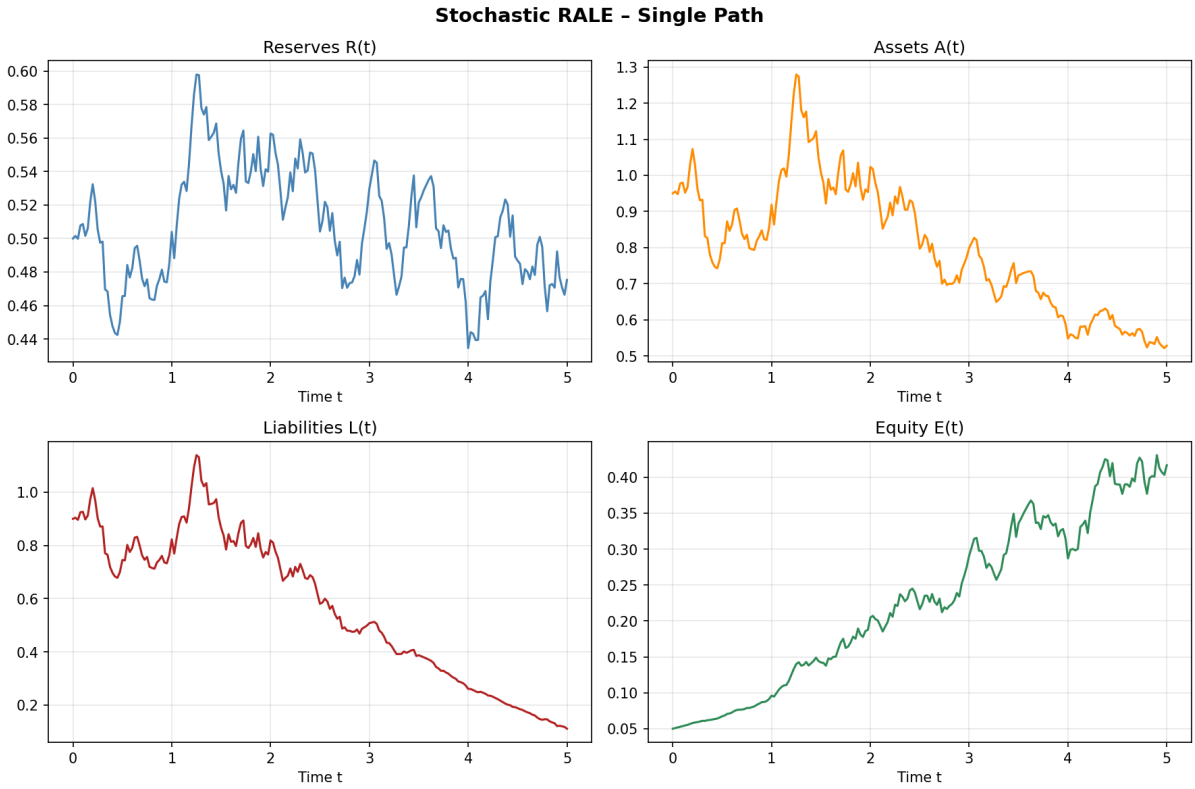

Figure 1: Single stochastic RALE path. The OU reserve process propagates uncertainty into all four balance-sheet variables. The equity curve's acceleration reflects path-dependent compounding.

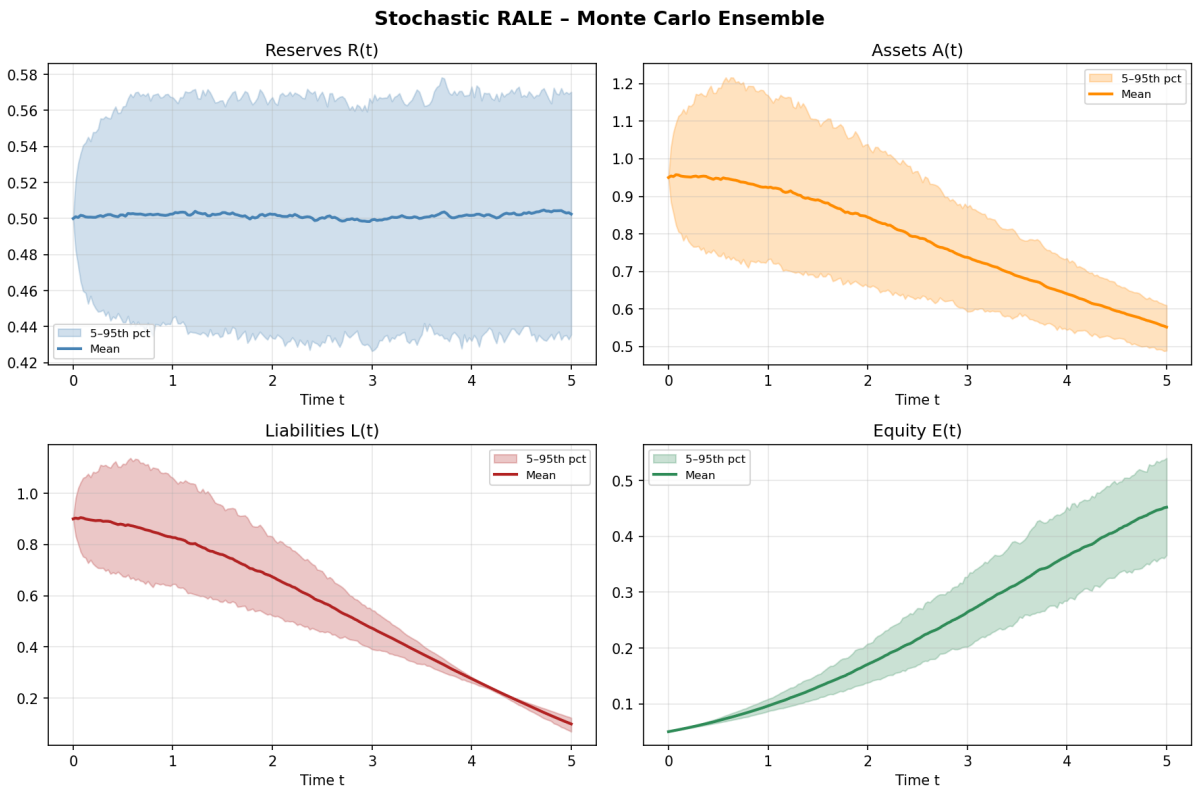

Figure 2: Monte Carlo ensemble (500 paths). Shaded bands show the 5th–95th percentile range; solid lines show the cross-sectional mean. Equity exhibits a widening fan due to path-dependent compounding.

7.2 Money Multiplier Sensitivity

Table 1 reports the mean effective money multiplier $m^*(T) = A^*(T)/R(T)$ at $T = 5$ across combinations of κ and σ, estimated from 200 Monte Carlo paths each.

Table 1: Effective money multiplier $m^*(T)$ at $T = 5$ across OU parameters (200 paths per cell). Higher σ and lower κ widen the confidence band and increase systemic leverage.

κ	σ	Mean $m^*(T)$	5th pct	95th pct
1.0	0.04	0.926	0.876	0.975
1.0	0.08	0.930	0.826	1.037
1.0	0.14	0.936	0.764	1.126
2.0	0.04	0.918	0.884	0.952
2.0	0.08	0.924	0.853	0.998
2.0	0.14	0.932	0.801	1.071
4.0	0.04	0.913	0.889	0.937
4.0	0.08	0.916	0.868	0.963
4.0	0.14	0.922	0.831	1.016

7.3 Risk Metric Trajectories

Figure 3 shows the time evolution of the insolvency probability $\mathbb{P}(E(t) \leq 0.01)$ and the reserve-floor breach probability $\mathbb{P}(R(t) \leq 0.05)$ across the 500-path ensemble.

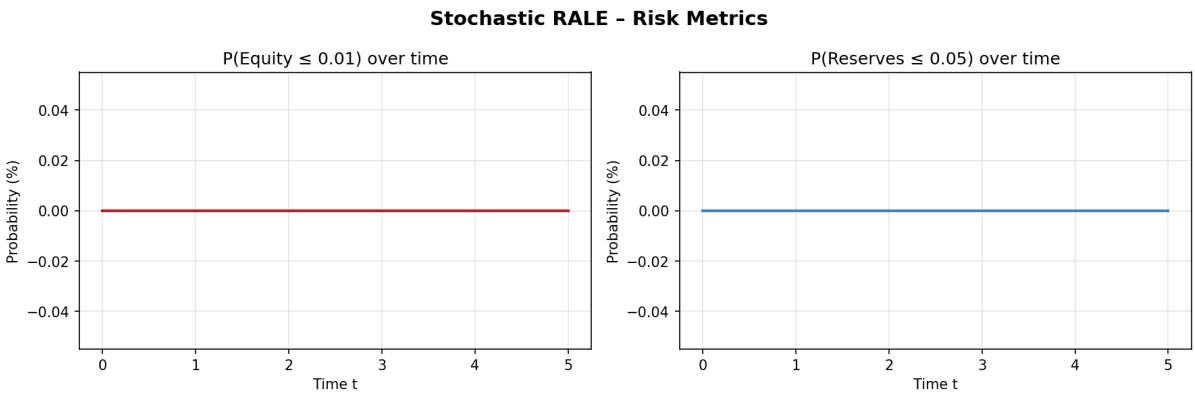

Figure 3: Risk metric trajectories. Both probabilities are initially low and rise over time as the OU process spreads, illustrating the gradual build-up of tail risk under stochastic reserves.

8 Conclusion

We have extended the RALE model to incorporate stochastic mean-reverting reserves modelled as an Ornstein–Uhlenbeck diffusion. The key contributions are:

1. **Almost-sure existence and uniqueness** of the stochastic RALE trajectory (Theorem 3.1), carrying over the deterministic result path-by-path via the implicit function theorem.
2. **Stationary multiplier distribution** characterised via a delta-method approximation (7)–(8), explaining the observed right-skewness of bank leverage ratios.
3. **Closed-form risk metrics** for insolvency (10) and reserve-floor breach (11) probabilities, expressed in terms of the Gaussian CDF Φ.
4. **Exact OU discretisation** (equation (12)) combined with the bisection RALE solver, achieving machine precision in under 34 inner iterations per time step.

Planned future directions include: (i) correlated multi-bank OU processes for the interbank contagion extension; (ii) jump-diffusion reserves to model emergency liquidity operations; and (iii) embedding the stochastic RALE within a DSGE general-equilibrium framework with monetary policy feedback.

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Glossary of Notation

Symbol	Meaning
$R(t)$	Reserve ratio at time t
$A(t)$	Total assets at time t
$L(t)$	Total liabilities at time t
$E(t)$	Equity (shareholders' residual claim) at time t
E_0	Initial equity level
$m(t)$	Effective money multiplier $A(t)/R(t)$
κ	OU mean-reversion speed
μ_R	OU long-run mean (policy target)
σ	OU volatility
$W(t)$	Standard Brownian motion
Φ	Standard normal CDF
Ω_{good}	Event that $R(t) \in (0,1)$ for all $t \in [0,T]$
$A^*(t)$	Unique solution of the RALE fixed-point equation
$G(R,t)$	Smooth function mapping $(R,t) \mapsto A^*$
R_{floor}	Regulatory reserve floor
ε_E	Insolvency equity threshold

The End